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Quadratic Equation and Inequalities

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	175 questions
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MCQ (Single Correct Answer) 175 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

Let $\lim\limits_{x\to 2}\frac{(\tan(x-2))(rx^2+(p-2)x-2p)}{(x-2)^2}=5$ for some $r, p \in \boldsymbol{R}$. If the set of all possible values of q , such that the roots of the equation $rx^2-px+q=0$ lie in $(0,2)$, be the interval $(\alpha,\beta]$, then $4(\alpha+\beta)$ equals :

(A) 11 (B) 13 (C) 17 (D) 21

Open solution

Q2. JEE Main 2026 (Online) 4th April Evening Shift

If the quadratic equation $(\lambda+2)x^2-3\lambda x+4\lambda=0, \lambda\neq -2$, has two positive roots, then the number of possible integral values of λ is :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q3. JEE Main 2026 (Online) 4th April Morning Shift

If the set of all solutions of $\left|x^2+x-9\right|=|x|+\left|x^2-9\right|$ is $[\alpha,\beta]\cup[\gamma,\infty)$, then ($\alpha^2+\beta^2+\gamma^2$) is equal to:

(A) 9 (B) 18 (C) 36 (D) 72

Open solution

Q4. JEE Main 2026 (Online) 2nd April Evening Shift

Let α, β be the roots of the equation $x^2-3x+r=0$, and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2+3x+r=0$.

If the roots of the equation $x^2+6x=m$ are $2\alpha+\beta+2r$ and $\alpha-2\beta-\frac{r}{2}$, then m is equal to :

(A) -135 (B) -567 (C) 135 (D) 567

Open solution

Q5. JEE Main 2026 (Online) 2nd April Morning Shift

Let $\alpha, \alpha+2, \alpha\in \mathbf{Z}$, be the roots of the quadratic equation $x(x+2)+(x+1)(x+3)+(x+2)(x+4)+\ldots.+(x+n-1)(x+n+1)=4n$ for some $n\in \mathbf{N}$. Then $n+\alpha$ is equal to :

(A) 0 (B) 1 (C) 2 (D) 3

Open solution

Q6. JEE Main 2026 (Online) 28th January Morning Shift

If α, β, where $\alpha<\beta$, are the roots of the equation $\lambda x^2-(\lambda+3)x+3=0$ such that $\frac{1}{\alpha}-\frac{1}{\beta}=\frac{1}{3}$, then the sum of all possible values of λ is

(A) 2 (B) 6 (C) 8 (D) 4

Open solution

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Q7. JEE Main 2026 (Online) 28th January Morning Shift

Let $S = \{x^3 + ax^2 + bx + c : a, b, c \in N \text{ and } a, b, c \leq 20\}$ be a set of polynomials. Then the number of polynomials in S , which are divisible by $x^2 + 2$, is

(A) 6 (B) 120 (C) 20 (D) 10

Open solution

Q8. JEE Main 2026 (Online) 24th January Evening Shift

The smallest positive integral value of a, for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

(A) 7 (B) 3 (C) 4 (D) 9

Open solution

Q9. JEE Main 2026 (Online) 24th January Morning Shift

The number of the real solutions of the equation: $x|x + 3| + |x - 1| - 2 = 0$ is

(A) 3 (B) 2 (C) 5 (D) 4

Open solution

Q10. JEE Main 2026 (Online) 23rd January Morning Shift

If α and $\beta(\alpha < \beta)$ are the roots of the equation
$(-2 + \sqrt{3})(|\sqrt{x} - 3|) + (x - 6\sqrt{x}) + (9 - 2\sqrt{3}) = 0, x \geqslant 0$, then $\sqrt{\frac{\beta}{\alpha}} + \sqrt{\alpha\beta}$ is equal to :

(A) 8 (B) 10 (C) 9 (D) 11

Open solution

Q11. JEE Main 2026 (Online) 23rd January Morning Shift

A building construction work can be completed by two masons A and B together in 22.5 days. Mason A alone can complete the construction work in 24 days less than mason B alone. Then mason A alone will complete the construction work in :

(A) 36 days (B) 24 days (C) 30 days (D) 42 days

Open solution

Q12. JEE Main 2026 (Online) 22nd January Evening Shift

Let α, β be the roots of the quadratic equation $12x^2 - 20x + 3\lambda = 0, \lambda \in \mathbf{Z}$. If $\frac{1}{2} \leqslant |\beta - \alpha| \leqslant \frac{3}{2}$, then the sum of all possible values of λ is :

(A) 3 (B) 6 (C) 1 (D) 4

Open solution

Q13. JEE Main 2026 (Online) 22nd January Morning Shift

The number of distinct real solutions of the equation $x|x + 4| + 3|x + 2| + 10 = 0$ is

(A) 2 (B) 3 (C) 0 (D) 1

Open solution

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Q14. JEE Main 2026 (Online) 21st January Evening Shift

Let α and β be the roots of the equation $x^2 + 2ax + (3a + 10) = 0$ such that $\alpha < 1 < \beta$. Then the set of all possible values of a is :

(A) $\left(-\infty, \frac{-11}{5}\right) \cup (5, \infty)$
(B) $\left(-\infty, \frac{-11}{5}\right)$
(C) $(-\infty, -3)$
(D) $(-\infty, -2) \cup (5, \infty)$

Open solution

Q15. JEE Main 2026 (Online) 21st January Morning Shift

The sum of all the roots of the equation $(x-1)^2 - 5|x-1| + 6 = 0$, is :

(A) 3
(B) 1
(C) 4
(D) 5

Open solution

Q16. JEE Main 2025 (Online) 8th April Evening Shift

The sum of the squares of the roots of $|x-2|^2 + |x-2| - 2 = 0$ and the squares of the roots of $x^2 - 2|x-3| - 5 = 0$, is

(A) 24
(B) 26
(C) 36
(D) 30

Open solution

Q17. JEE Main 2025 (Online) 7th April Evening Shift

The number of real roots of the equation $x|x-2| + 3|x-3| + 1 = 0$ is :

(A) 4
(B) 3
(C) 2
(D) 1

Open solution

Q18. JEE Main 2025 (Online) 7th April Morning Shift

Let the set of all values of $p \in \mathbb{R}$, for which both the roots of the equation $x^2 - (p+2)x + (2p+9) = 0$ are negative real numbers, be the interval $(\alpha, \beta]$. Then $\beta - 2\alpha$ is equal to

(A) 5
(B) 0
(C) 20
(D) 9

Open solution

Q19. JEE Main 2025 (Online) 4th April Morning Shift

Consider the equation $x^2 + 4x - n = 0$, where $n \in [20, 100]$ is a natural number. Then the number of all distinct values of n, for which the given equation has integral roots, is equal to

(A) 6
(B) 5
(C) 8
(D) 7

Open solution

Q20. JEE Main 2025 (Online) 3rd April Evening Shift

Let the equation $x(x+2)(12-k) = 2$ have equal roots. Then the distance of the point $\left(k, \frac{k}{2}\right)$ from the line $3x + 4y + 5 = 0$ is

(A) 15
(B) 12
(C) $5\sqrt{3}$
(D) $15\sqrt{5}$

Open solution

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Q21. JEE Main 2025 (Online) 3rd April Morning Shift

Let α and β be the roots of $x^2+\sqrt{3}x-16=0$, and γ and δ be the roots of $x^2+3x-1=0$. If $P_n=\alpha^n+\beta^n$ and $Q_n=\gamma^n+\delta^n$, then $\frac{P_{25}+\sqrt{3}P_{24}}{2P_{23}}+\frac{Q_{25}-Q_{23}}{Q_{24}}$ is equal to

(A) 4 (B) 3 (C) 5 (D) 7

Open solution

Q22. JEE Main 2025 (Online) 2nd April Morning Shift

Let $\mathrm{P_n}=\alpha^{\mathrm{n}}+\beta^{\mathrm{n}}, \mathrm{n}\in \mathrm{N}$. If $\mathrm{P_{10}}=123, \mathrm{P_9}=76, \mathrm{P_8}=47$ and $\mathrm{P_1}=1$, then the quadratic equation having roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is :

(A) $x^2+x-1=0$ (B) $x^2-x+1=0$ (C) $x^2+x+1=0$ (D) $x^2-x-1=0$

Open solution

Q23. JEE Main 2025 (Online) 29th January Evening Shift

If the set of all $a\in\mathbf{R}$, for which the equation $2x^2+(a-5)x+15=3a$ has no real root, is the interval (α,β), and $X=\{x\in Z;\alpha<x<\beta\}$, then $\sum_{x\in X}x^2$ is equal to:

(A) 2139 (B) 2119 (C) 2109 (D) 2129

Open solution

Q24. JEE Main 2025 (Online) 29th January Morning Shift

The number of solutions of the equation

$\left(\frac{9}{x}-\frac{9}{\sqrt{x}}+2\right)\left(\frac{2}{x}-\frac{7}{\sqrt{x}}+3\right)=0$ is :

(A) 3 (B) 2 (C) 1 (D) 4

Open solution

Q25. JEE Main 2025 (Online) 28th January Evening Shift

Let $f:\mathbf{R}-\{0\}\to(-\infty,1)$ be a polynomial of degree 2 , satisfying $f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$. If $f(\mathrm{K})=-2\,\mathrm{K}$, then the sum of squares of all possible values of K is :

(A) 9 (B) 1 (C) 6 (D) 7

Open solution

Q26. JEE Main 2025 (Online) 28th January Morning Shift

The sum, of the squares of all the roots of the equation $x^2+|2x-3|-4=0$, is

(A) $6(2-\sqrt{2})$ (B) $3(3-\sqrt{2})$ (C) $3(2-\sqrt{2})$ (D) $6(3-\sqrt{2})$

Open solution

Q27. JEE Main 2025 (Online) 24th January Evening Shift

The number of real solution(s) of the equation $x^2+3x+2=\min\{|x-3|,|x+2|\}$ is :

(A) 2 (B) 3 (C) 1 (D) 0

Open solution

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Q28. JEE Main 2025 (Online) 24th January Morning Shift

The product of all the rational roots of the equation $\left(x^2-9x+11\right)^2-(x-4)(x-5)=3$, is equal to

(A) 7 (B) 21 (C) 28 (D) 14

Open solution

Q29. JEE Main 2025 (Online) 22nd January Evening Shift

Let α_θ and β_θ be the distinct roots of $2x^2+(\cos\theta)x-1=0, \theta\in(0,2\pi)$. If m and M are the minimum and the maximum values of $\alpha_\theta^4+\beta_\theta^4$, then $16(M+m)$ equals :

(A) 27 (B) 17 (C) 25 (D) 24

Open solution

Q30. JEE Main 2024 (Online) 9th April Evening Shift

Let $\alpha,\beta;\alpha>\beta$, be the roots of the equation $x^2-\sqrt{2}x-\sqrt{3}=0$. Let $\mathrm{P}_n=\alpha^n-\beta^n, n\in\mathrm{N}$. Then $(11\sqrt{3}-10\sqrt{2})\mathrm{P}_{10}+(11\sqrt{2}+10)\mathrm{P}_{11}-11\mathrm{P}_{12}$ is equal to

(A) $10\sqrt{3}\mathrm{P}_9$ (B) $11\sqrt{3}\mathrm{P}_9$ (C) $11\sqrt{2}\mathrm{P}_9$ (D) $10\sqrt{2}\mathrm{P}_9$

Open solution

Q31. JEE Main 2024 (Online) 9th April Morning Shift

Let α,β be the roots of the equation $x^2+2\sqrt{2}x-1=0$. The quadratic equation, whose roots are $\alpha^4+\beta^4$ and $\frac{1}{10}(\alpha^6+\beta^6)$, is:

(A) $x^2-180x+9506=0$ (B) $x^2-195x+9506=0$

(C) $x^2-190x+9466=0$ (D) $x^2-195x+9466=0$

Open solution

Q32. JEE Main 2024 (Online) 8th April Morning Shift

The sum of all the solutions of the equation $(8)^{2x}-16\cdot(8)^x+48=0$ is :

(A) $1+\log_8(6)$ (B) $1+\log_6(8)$ (C) $\log_8(6)$ (D) $\log_8(4)$

Open solution

Q33. JEE Main 2024 (Online) 6th April Morning Shift

Let α,β be the distinct roots of the equation $x^2-\left(t^2-5t+6\right)x+1=0, t\in\mathbb{R}$ and $a_n=\alpha^n+\beta^n$. Then the minimum value of $\frac{a_{2023}+a_{2025}}{a_{2024}}$ is

(A) $-1/2$ (B) $-1/4$ (C) $1/4$ (D) $1/2$

Open solution

Q34. JEE Main 2024 (Online) 4th April Morning Shift

If 2 and 6 are the roots of the equation $ax^2+bx+1=0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is :

(A) $x^2+8x+12=0$ (B) $2x^2+11x+12=0$

(C) $4x^2+14x+12=0$ (D) $x^2+10x+16=0$

Open solution

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Q35. JEE Main 2024 (Online) 1st February Evening Shift

Let α and β be the roots of the equation $px^2 + qx - r = 0$, where $p \neq 0$. If p, q and r be the consecutive terms of a non constant G.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{3}{4}$, then the value of $(\alpha - \beta)^2$ is :

(A) 8 (B) 9 (C) $\frac{20}{3}$ (D) $\frac{80}{9}$

Open solution

Q36. JEE Main 2024 (Online) 1st February Morning Shift

Let $\mathbf{S} = \left\{x \in \mathbf{R} : (\sqrt{3}+\sqrt{2})^x + (\sqrt{3}-\sqrt{2})^x = 10\right\}$. Then the number of elements in S is :

(A) 4 (B) 0 (C) 2 (D) 1

Open solution

Q37. JEE Main 2024 (Online) 31st January Morning Shift

Let S be the set of positive integral values of a for which $\frac{ax^2+2(a+1)x+9a+4}{x^2-8x+32} < 0, \forall x \in \mathbb{R}$. Then, the number of elements in S is :

(A) 0 (B) ∞ (C) 3 (D) 1

Open solution

Q38. JEE Main 2024 (Online) 27th January Evening Shift

If α, β are the roots of the equation, $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then :

(A) $2S_{12} = S_{11} + S_{10}$ (B) $S_{12} = S_{11} + S_{10}$ (C) $S_{11} = S_{10} + S_{12}$ (D) $2S_{11} = S_{12} + S_{10}$

Open solution

Q39. JEE Main 2023 (Online) 15th April Morning Shift

The number of real roots of the equation $x|x| - 5|x+2| + 6 = 0$, is :

(A) 4 (B) 3 (C) 5 (D) 6

Open solution

Q40. JEE Main 2023 (Online) 13th April Evening Shift

Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$. Then $\alpha^{14} + \beta^{14}$ is equal to

(A) -64 (B) $-64\sqrt{2}$ (C) $-128\sqrt{2}$ (D) -128

Open solution

Q41. JEE Main 2023 (Online) 13th April Morning Shift

The set of all $a \in \mathbb{R}$ for which the equation $x|x-1| + |x+2| + a = 0$ has exactly one real root, is :

(A) $(-\infty, \infty)$ (B) $(-6, \infty)$ (C) $(-\infty, -3)$ (D) $(-6, -3)$

Open solution

Q42. JEE Main 2023 (Online) 12th April Morning Shift

Let α, β be the roots of the quadratic equation $x^2 + \sqrt{6}x + 3 = 0$. Then $\frac{\alpha^{23}+\beta^{23}+\alpha^{14}+\beta^{14}}{\alpha^{15}+\beta^{15}+\alpha^{10}+\beta^{10}}$ is equal to :

(A) 72 (B) 9 (C) 729 (D) 81

Open solution

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Q43. JEE Main 2023 (Online) 8th April Morning Shift

Let α, β, γ be the three roots of the equation $x^3 + bx + c = 0$. If $\beta\gamma = 1 = -\alpha$, then $b^3 + 2c^3 - 3\alpha^3 - 6\beta^3 - 8\gamma^3$ is equal to :

(A) 21 (B) 19 (C) $\frac{169}{8}$ (D) $\frac{155}{8}$

Open solution

Q44. JEE Main 2023 (Online) 6th April Morning Shift

Let $A = \{x \in R : [x + 3] + [x + 4] \leq 3\}$,

$B = \left\{x \in R : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r}\right)^{x-3} < 3^{-3x}\right\}$, where [t] denotes greatest integer function. Then,

(A) $B \subset C, A \neq B$ (B) $A \subset B, A \neq B$ (C) $A = B$ (D) $A \cap B = \phi$

Open solution

Q45. JEE Main 2023 (Online) 6th April Morning Shift

The sum of all the roots of the equation $|x^2 - 8x + 15| - 2x + 7 = 0$ is :

(A) $11 + \sqrt{3}$ (B) $9 + \sqrt{3}$ (C) $9 - \sqrt{3}$ (D) $11 - \sqrt{3}$

Open solution

Q46. JEE Main 2023 (Online) 1st February Evening Shift

The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval (1, 2) and its other root lies in the interval (2, 3), is :

(A) 2 (B) 0 (C) 1 (D) 3

Open solution

Q47. JEE Main 2023 (Online) 1st February Morning Shift

Let $S = \left\{x : x \in \mathbb{R} \text{ and } (\sqrt{3} + \sqrt{2})^{x^2-4} + (\sqrt{3} - \sqrt{2})^{x^2-4} = 10\right\}$. Then $n(S)$ is equal to

(A) 6 (B) 4 (C) 0 (D) 2

Open solution

Q48. JEE Main 2023 (Online) 31st January Evening Shift

The equation $e^{4x} + 8e^{3x} + 13e^{2x} - 8e^x + 1 = 0, x \in \mathbb{R}$ has :

(A) two solutions and both are negative

(B) two solutions and only one of them is negative

(C) four solutions two of which are negative

(D) no solution

Open solution

Q49. JEE Main 2023 (Online) 31st January Morning Shift

The number of real roots of the equation $\sqrt{x^2 - 4x + 3} + \sqrt{x^2 - 9} = \sqrt{4x^2 - 14x + 6}$, is :

(A) 0 (B) 1 (C) 3 (D) 2

Open solution

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Q50. JEE Main 2023 (Online) 29th January Morning Shift

Let $\lambda \neq 0$ be a real number. Let α, β be the roots of the equation $14x^2 - 31x + 3\lambda = 0$ and α, γ be the roots of the equation $35x^2 - 53x + 4\lambda = 0$. Then $\frac{3\alpha}{\beta}$ and $\frac{4\alpha}{\gamma}$ are the roots of the equation

(A) $7x^2 - 245x + 250 = 0$
(B) $49x^2 - 245x + 250 = 0$
(C) $49x^2 + 245x + 250 = 0$
(D) $7x^2 + 245x - 250 = 0$

Open solution

Q51. JEE Main 2023 (Online) 24th January Evening Shift

The number of real solutions of the equation $3\left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) + 5 = 0$, is

(A) 3
(B) 4
(C) 0
(D) 2

Open solution

Q52. JEE Main 2023 (Online) 24th January Morning Shift

The equation $x^2 - 4x + [x] + 3 = x[x]$, where $[x]$ denotes the greatest integer function, has :

(A) exactly two solutions in $(-\infty, \infty)$
(B) no solution
(C) a unique solution in $(-\infty, \infty)$
(D) a unique solution in $(-\infty, 1)$

Open solution

Q53. JEE Main 2022 (Online) 28th July Evening Shift

Let $S = \left\{x \in [-6, 3] - \{-2, 2\} : \frac{|x+3|-1}{|x|-2} \geq 0\right\}$ and $T = \left\{x \in \mathbb{Z} : x^2 - 7|x| + 9 \leq 0\right\}$.

Then the number of elements in $S \cap T$ is :

(A) 7
(B) 5
(C) 4
(D) 3

Open solution

Q54. JEE Main 2022 (Online) 28th July Evening Shift

Let α, β be the roots of the equation $x^2 - \sqrt{2}x + \sqrt{6} = 0$ and $\frac{1}{\alpha^2} + 1, \frac{1}{\beta^2} + 1$ be the roots of the equation $x^2 + ax + b = 0$. Then the roots of the equation $x^2 - (a + b - 2)x + (a + b + 2) = 0$ are :

(A) non-real complex numbers
(B) real and both negative
(C) real and both positive
(D) real and exactly one of them is positive

Open solution

Q55. JEE Main 2022 (Online) 27th July Evening Shift

If α, β are the roots of the equation

$$x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right)x + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0,$$

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, is :

(A) $3x^2 - 20x - 12 = 0$
(B) $3x^2 - 10x - 4 = 0$
(C) $3x^2 - 10x + 2 = 0$
(D) $3x^2 - 20x + 16 = 0$

Open solution

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Q56. JEE Main 2022 (Online) 26th July Evening Shift

The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is:

(A) 4 (B) 5 (C) 6 (D) 8

Open solution

Q57. JEE Main 2022 (Online) 25th July Morning Shift

If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 + x^3 + x^2 + x + 1 = 0$, then $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021}$ is equal to :

(A) -4 (B) -1 (C) 1 (D) 4

Open solution

Q58. JEE Main 2022 (Online) 30th June Morning Shift

Let $S_1 = \left\{x \in R - \{1, 2\} : \frac{(x+2)(x^2+3x+5)}{-2+3x-x^2} \geq 0\right\}$ and $S_2 = \left\{x \in R : 3^{2x} - 3^{x+1} - 3^{x+2} + 27 \leq 0\right\}$. Then, $S_1 \cup S_2$ is equal to :

(A) $(-\infty, -2] \cup (1, 2)$ (B) $(-\infty, -2] \cup [1, 2]$ (C) $(-2, 1] \cup [2, \infty)$ (D) $(-\infty, 2]$

Open solution

Q59. JEE Main 2022 (Online) 30th June Morning Shift

Let S be the set of all integral values of α for which the sum of squares of two real roots of the quadratic equation $3x^2 + (\alpha - 6)x + (\alpha + 3) = 0$ is minimum. Then S :

(A) is an empty set (B) is a singleton

(C) contains exactly two elements (D) contains more than two elements

Open solution

Q60. JEE Main 2022 (Online) 29th June Evening Shift

Let α be a root of the equation $1 + x^2 + x^4 = 0$. Then, the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to :

(A) 1 (B) α (C) $1 + \alpha$ (D) $1 + 2\alpha$

Open solution

Q61. JEE Main 2022 (Online) 28th June Evening Shift

Let f(x) be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1, then the sum of the roots of $f(x) = 0$ is equal to :

(A) $\frac{11}{3}$ (B) $\frac{7}{3}$ (C) $\frac{13}{3}$ (D) $\frac{14}{3}$

Open solution

Q62. JEE Main 2022 (Online) 27th June Morning Shift

The number of distinct real roots of $x^4 - 4x + 1 = 0$ is :

(A) 4 (B) 2 (C) 1 (D) 0

Open solution

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Q63. JEE Main 2022 (Online) 25th June Evening Shift

Let $A = \{x \in R : |x + 1| < 2\}$ and $B = \{x \in R : |x - 1| \geq 2\}$. Then which one of the following statements is NOT true?

(A) $A - B = (-1, 1)$
(B) $B - A = R - (-3, 1)$
(C) $A \cap B = (-3, -1]$
(D) $A \cup B = R - [1, 3)$

Open solution

Q64. JEE Main 2022 (Online) 25th June Evening Shift

Let a, b ∈ R be such that the equation $ax^2 - 2bx + 15 = 0$ has a repeated root α. If α and β are the roots of the equation $x^2 - 2bx + 21 = 0$, then $\alpha^2 + \beta^2$ is equal to :

(A) 37
(B) 58
(C) 68
(D) 92

Open solution

Q65. JEE Main 2022 (Online) 24th June Evening Shift

The sum of all the real roots of the equation

$(e^{2x} - 4)(6e^{2x} - 5e^x + 1) = 0$ is

(A) $\log_e 3$
(B) $-\log_e 3$
(C) $\log_e 6$
(D) $-\log_e 6$

Open solution

Q66. JEE Main 2022 (Online) 24th June Evening Shift

The number of distinct real roots of the equation

$x^7 - 7x - 2 = 0$ is

(A) 5
(B) 7
(C) 1
(D) 3

Open solution

Q67. JEE Main 2022 (Online) 24th June Morning Shift

If the sum of the squares of the reciprocals of the roots α and β of

the equation $3x^2 + \lambda x - 1 = 0$ is 15, then $6(\alpha^3 + \beta^3)^2$ is equal to :

(A) 18
(B) 24
(C) 36
(D) 96

Open solution

Q68. JEE Main 2021 (Online) 1st September Evening Shift

The number of pairs (a, b) of real numbers, such that whenever α is a root of the equation $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root of this equation, is :

(A) 6
(B) 2
(C) 4
(D) 8

Open solution

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Q69. JEE Main 2021 (Online) 31st August Evening Shift

The sum of the roots of the equation

$x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 0$, is :

(A) $\log_2 14$ (B) $\log_2 11$ (C) $\log_2 12$ (D) $\log_2 13$

Open solution

Q70. JEE Main 2021 (Online) 31st August Morning Shift

cosec18° is a root of the equation :

(A) $x^2 + 2x - 4 = 0$ (B) $4x^2 + 2x - 1 = 0$ (C) $x^2 - 2x + 4 = 0$ (D) $x^2 - 2x - 4 = 0$

Open solution

Q71. JEE Main 2021 (Online) 27th August Evening Shift

The set of all values of $k > -1$, for which the equation $(3x^2 + 4x + 3)^2 - (k+1)(3x^2 + 4x + 3)(3x^2 + 4x + 2) + k(3x^2 + 4x + 2)^2 = 0$ has real roots, is :

(A) $\left(1, \frac{5}{2}\right]$ (B) [2, 3)

(C) $\left[-\frac{1}{2}, 1\right)$ (D) $\left(\frac{1}{2}, \frac{3}{2}\right] - \{1\}$

Open solution

Q72. JEE Main 2021 (Online) 27th July Evening Shift

Let $\alpha = \max_{x \in R}\{8^{2\sin 3x}.4^{4\cos 3x}\}$ and $\beta = \min_{x \in R}\{8^{2\sin 3x}.4^{4\cos 3x}\}$. If $8x^2 + bx + c = 0$ is a quadratic equation whose roots are $\alpha^{1/5}$ and $\beta^{1/5}$, then the value of c − b is equal to :

(A) 42 (B) 47 (C) 43 (D) 50

Open solution

Q73. JEE Main 2021 (Online) 27th July Morning Shift

Let α, β be two roots of the

equation $x^2 + (20)^{1/4}x + (5)^{1/2} = 0$. Then $\alpha^8 + \beta^8$ is equal to

(A) 10 (B) 100 (C) 50 (D) 160

Open solution

Q74. JEE Main 2021 (Online) 25th July Evening Shift

If [x] be the greatest integer less than or equal to x,

then $\sum_{n=8}^{100}\left[\frac{(-1)^n n}{2}\right]$ is equal to :

(A) 0 (B) 4 (C) −2 (D) 2

Open solution

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Q75. JEE Main 2021 (Online) 25th July Evening Shift

The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is :

(A) 2 (B) 3 (C) 1 (D) 4

Open solution

Q76. JEE Main 2021 (Online) 25th July Morning Shift

The number of real roots of the equation $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ is :

(A) 2 (B) 4 (C) 6 (D) 1

Open solution

Q77. JEE Main 2021 (Online) 22th July Evening Shift

Let [x] denote the greatest integer less than or equal to x. Then, the values of $x \in R$ satisfying the equation $[e^x]^2 + [e^x + 1] - 3 = 0$ lie in the interval :

(A) $\left[0, \frac{1}{e}\right)$ (B) [$\log_e 2$, $\log_e 3$) (C) [1, e) (D) [0, $\log_e 2$)

Open solution

Q78. JEE Main 2021 (Online) 20th July Morning Shift

If α and β are the distinct roots of the equation $x^2 + (3)^{1/4}x + 3^{1/2} = 0$, then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to :

(A) 56×3^{25} (B) 56×3^{24} (C) 52×3^{24} (D) 28×3^{25}

Open solution

Q79. JEE Main 2021 (Online) 18th March Morning Shift

The value of $3 + \cfrac{1}{4+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{3+....\infty}}}}$ is equal to

(A) $1.5 + \sqrt{3}$ (B) $2 + \sqrt{3}$ (C) $3 + 2\sqrt{3}$ (D) $4 + \sqrt{3}$

Open solution

Q80. JEE Main 2021 (Online) 17th March Morning Shift

The value of $4 + \cfrac{1}{5+\cfrac{1}{4+\cfrac{1}{5+\cfrac{1}{4+.....\infty}}}}$ is :

(A) $2 + \frac{2}{5}\sqrt{30}$ (B) $2 + \frac{4}{\sqrt{5}}\sqrt{30}$

(C) $5 + \frac{2}{5}\sqrt{30}$ (D) $4 + \frac{4}{\sqrt{5}}\sqrt{30}$

Open solution

Q81. JEE Main 2021 (Online) 16th March Morning Shift

The number of elements in the set $\{x \in R : (|x| - 3)|x + 4| = 6\}$ is equal to :

(A) 4 (B) 2 (C) 3 (D) 1

Open solution

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Q82. JEE Main 2021 (Online) 25th February Evening Shift

Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10}-2a_8}{3a_9}$ is :

(A) 3 (B) 2 (C) 4 (D) 1

Open solution

Q83. JEE Main 2021 (Online) 25th February Morning Shift

The integer 'k', for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in R, is :

(A) 4 (B) 2 (C) 3 (D) 0

Open solution

Q84. JEE Main 2021 (Online) 24th February Morning Shift

Let p and q be two positive numbers such that p + q = 2 and $p^4+q^4 = 272$. Then p and q are roots of the equation :

(A) $x^2 - 2x + 8 = 0$ (B) $x^2 - 2x + 136 = 0$ (C) $x^2 - 2x + 16 = 0$ (D) $x^2 - 2x + 2 = 0$

Open solution

Q85. JEE Main 2020 (Online) 6th September Evening Slot

If α and β are the roots of the equation $2x(2x + 1) = 1$, then β is equal to :

(A) $-2\alpha(\alpha + 1)$ (B) $2\alpha(\alpha + 1)$

(C) $2\alpha^2$ (D) $2\alpha(\alpha - 1)$

Open solution

Q86. JEE Main 2020 (Online) 6th September Morning Slot

If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{1/8} + \left(\frac{\beta^3}{\alpha^5}\right)^{1/8}$ is :

(A) 1 (B) 3 (C) 2 (D) 4

Open solution

Q87. JEE Main 2020 (Online) 5th September Evening Slot

If α and β are the roots of the equation, $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ is equal to :

(A) $\frac{1}{24}$ (B) $\frac{27}{32}$ (C) $\frac{27}{16}$ (D) $\frac{3}{8}$

Open solution

Q88. JEE Main 2020 (Online) 5th September Morning Slot

The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$ is :

(A) $\frac{5}{9}$ (B) $\frac{5}{27}$ (C) $\frac{25}{81}$ (D) $\frac{25}{9}$

Open solution

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Q89. JEE Main 2020 (Online) 4th September Evening Slot

Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to:

Open solution

(A) 36 (B) 9 (C) 27 (D) 18

Q90. JEE Main 2020 (Online) 4th September Morning Slot

Let [t] denote the greatest integer $\leq$ t. Then the equation in x, $[x]^2 + 2[x+2] - 7 = 0$ has :

Open solution

(A) no integral solution.
(B) exactly two solutions.
(C) exactly four integral solutions.
(D) infinitely many solutions.

Q91. JEE Main 2020 (Online) 4th September Morning Slot

Let α and β be the roots of $x^2 - 3x + p=0$ and γ and δ be the roots of $x^2 - 6x + q = 0$. If $\alpha, \beta, \gamma, \delta$ form a geometric progression.Then ratio (2q + p) : (2q - p) is:

Open solution

(A) 9 : 7 (B) 5 : 3 (C) 3 : 1 (D) 33 :31

Q92. JEE Main 2020 (Online) 3rd September Evening Slot

The set of all real values of λ for which the quadratic equations, $(\lambda^2 + 1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval (0, 1) is :

Open solution

(A) (−3, −1) (B) (2, 4] (C) (0, 2) (D) (1, 3]

Q93. JEE Main 2020 (Online) 3rd September Morning Slot

If α and β are the roots of the equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2qx + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to :

Open solution

(A) $\frac{9}{4}\left(9 - q^2\right)$ (B) $\frac{9}{4}\left(9 + q^2\right)$
(C) $\frac{9}{4}\left(9 - p^2\right)$ (D) $\frac{9}{4}\left(9 + p^2\right)$

Q94. JEE Main 2020 (Online) 2nd September Evening Slot

Let f(x) be a quadratic polynomial such that f(−1) + f(2) = 0. If one of the roots of f(x) = 0 is 3, then its other root lies in :

Open solution

(A) (−3, −1) (B) (1, 3) (C) (−1, 0) (D) (0, 1)

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Q95. JEE Main 2020 (Online) 2nd September Morning Slot

Let α and β be the roots of the equation
$5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n$, n = 1, 2, 3...., then :

(A) $5S_6 + 6S_5 = 2S_4$

(B) $5S_6 + 6S_5 + 2S_4 = 0$

(C) $6S_6 + 5S_5 + 2S_4 = 0$

(D) $6S_6 + 5S_5 = 2S_4$

Open solution

Q96. JEE Main 2020 (Online) 9th January Evening Slot

Let a, b $\in$ R, a $\neq$ 0 be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α, which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to :

(A) 28

(B) 24

(C) 26

(D) 25

Open solution

Q97. JEE Main 2020 (Online) 9th January Morning Slot

The number of real roots of the equation,
$e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is :

(A) 1

(B) 2

(C) 3

(D) 4

Open solution

Q98. JEE Main 2020 (Online) 8th January Evening Slot

Let S be the set of all real roots of the equation,
$3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$. Then S :

(A) contains exactly two elements.

(B) is an empty set.

(C) is a singleton.

(D) contains at least four elements.

Open solution

Q99. JEE Main 2020 (Online) 8th January Evening Slot

Let $\alpha = \frac{-1+i\sqrt{3}}{2}$.
If $a = (1+\alpha)\sum_{k=0}^{100}\alpha^{2k}$ and
$b = \sum_{k=0}^{100}\alpha^{3k}$, then a and b are the roots of the quadratic equation :

(A) $x^2 + 101x + 100 = 0$

(B) $x^2 + 102x + 101 = 0$

(C) $x^2 - 102x + 101 = 0$

(D) $x^2 - 101x + 100 = 0$

Open solution

Q100. JEE Main 2020 (Online) 7th January Evening Slot

Let α and β be the roots of the equation $x^2 - x - 1 = 0$.
If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true?

(A) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$

(B) $p_5 = 11$

(C) $p_3 = p_5 - p_4$

(D) $p_5 = p_2 \cdot p_3$

Open solution

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Q101. JEE Main 2020 (Online) 7th January Morning Slot

Let α and β be two real roots of the equation
$(k + 1)\tan^2 x - \sqrt{2} \cdot \lambda \tan x = (1 - k)$, where $k(\neq -1)$ and λ are real numbers. if $\tan^2(\alpha + \beta) = 50$, then a value of λ is:

Open solution

(A) $5\sqrt{2}$ (B) 10 (C) 5 (D) $10\sqrt{2}$

Q102. JEE Main 2019 (Online) 12th April Evening Slot

If α, β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, then $\alpha(\beta + \gamma)$ is equal to :

Open solution

(A) $\alpha\gamma$ (B) 0 (C) $\beta\gamma$ (D) $\alpha\beta$

Q103. JEE Main 2019 (Online) 10th April Evening Slot

The number of real roots of the equation
$5 + |2^x - 1| = 2^x(2^x - 2)$ is

Open solution

(A) 2 (B) 1 (C) 3 (D) 4

Q104. JEE Main 2019 (Online) 10th April Morning Slot

All the pairs (x, y) that satisfy the inequality
$2^{\sqrt{\sin^2 x - 2\sin x + 5}} \cdot \frac{1}{4^{\sin^2 y}} \leq 1$
also satisfy the equation

Open solution

(A) $\sin x = |\sin y|$ (B) $\sin x = 2\sin y$
(C) $2\sin x = \sin y$ (D) $2|\sin x| = 3\sin y$

Q105. JEE Main 2019 (Online) 10th April Morning Slot

If α and β are the roots of the quadratic equation,
$x^2 + x\sin\theta - 2\sin\theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$, then
$\frac{\alpha^{12} + \beta^{12}}{(\alpha^{-12} + \beta^{-12}) \cdot (\alpha - \beta)^{24}}$ is equal to :

Open solution

(A) $\frac{2^{12}}{(\sin\theta - 8)^6}$ (B) $\frac{2^6}{(\sin\theta + 4)^{12}}$
(C) $\frac{2^{12}}{(\sin\theta + 8)^{12}}$ (D) $\frac{2^{12}}{(\sin\theta - 4)^{12}}$

Q106. JEE Main 2019 (Online) 9th April Evening Slot

If m is chosen in the quadratic equation

$(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$

such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is :-

Open solution

(A) $4\sqrt{3}$ (B) $8\sqrt{3}$ (C) $8\sqrt{5}$ (D) $10\sqrt{5}$

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Q107. JEE Main 2019 (Online) 9th April Morning Slot

Let p, q ∈ R. If $2 - \sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then :

(A) $p^2 - 4q - 12 = 0$ (B) $q^2 - 4p - 16 = 0$ (C) $q^2 + 4p + 14 = 0$ (D) $p^2 - 4q + 12 = 0$

Open solution

Q108. JEE Main 2019 (Online) 8th April Evening Slot

The number of integral values of m for which the equation

$(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is :

(A) 2 (B) infinitely many (C) 1 (D) 3

Open solution

Q109. JEE Main 2019 (Online) 8th April Morning Slot

The sum of the solutions of the equation

$|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0$

(x > 0) is equal to:

(A) 9 (B) 12 (C) 4 (D) 10

Open solution

Q110. JEE Main 2019 (Online) 12th January Evening Slot

The number of integral values of m for which the quadratic expression, $(1 + 2m)x^2 - 2(1 + 3m)x + 4(1 + m)$, $x \in R$, is always positive, is :

(A) 7 (B) 8 (C) 3 (D) 6

Open solution

Q111. JEE Main 2019 (Online) 12th January Morning Slot

If λ be the ratio of the roots of the quadratic equation in x, $3m^2x^2 + m(m - 4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is

(A) $-2 + \sqrt{2}$ (B) $4 - 3\sqrt{2}$ (C) $2 - \sqrt{3}$ (D) $4 - 2\sqrt{3}$

Open solution

Q112. JEE Main 2019 (Online) 11th January Evening Slot

Let α and β be the roots of the quadratic equation $x^2 \sin\theta - x(\sin\theta\cos\theta + 1) + \cos\theta = 0$ ($0 < \theta < 45^\circ$), and $\alpha < \beta$. Then $\sum_{n=0}^{\infty}\left(\alpha^n + \frac{(-1)^n}{\beta^n}\right)$ is equal to :

(A) $\frac{1}{1+\cos\theta} + \frac{1}{1-\sin\theta}$ (B) $\frac{1}{1-\cos\theta} + \frac{1}{1+\sin\theta}$

(C) $\frac{1}{1-\cos\theta} - \frac{1}{1+\sin\theta}$ (D) $\frac{1}{1+\cos\theta} - \frac{1}{1-\sin\theta}$

Open solution

Q113. JEE Main 2019 (Online) 11th January Morning Slot

If one real root of the quadratic equation $81x^2 + kx + 256 = 0$ is cube of the other root, then a value of k is

(A) -81 (B) -300 (C) 100 (D) 144

Open solution

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Q114. JEE Main 2019 (Online) 10th January Evening Slot

The value of λ such that sum of the squares of the roots of the quadratic equation, $x^2 + (3 - \lambda)x + 2 = \lambda$ has the least value is -

(A) 1 (B) 2 (C) $\frac{15}{8}$ (D) $\frac{4}{9}$

Open solution

Q115. JEE Main 2019 (Online) 10th January Morning Slot

If 5, 5r, $5r^2$ are the lengths of the sides of a triangle, then r cannot be equal to :

(A) $\frac{7}{4}$ (B) $\frac{5}{4}$ (C) $\frac{3}{4}$ (D) $\frac{3}{2}$

Open solution

Q116. JEE Main 2019 (Online) 10th January Morning Slot

Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval (0, 2) and its other root lies in the interval (2, 3). Then the number of elements in S is -

(A) 12 (B) 18 (C) 10 (D) 11

Open solution

Q117. JEE Main 2019 (Online) 9th January Evening Slot

If both the roots of the quadratic equation $x^2 - mx + 4 = 0$ are real and distinct and they lie in the interval [1, 5], then m lies in the interval :

(A) $(-5, -4)$ (B) (4, 5) (C) (5, 6) (D) (3, 4)

Open solution

Q118. JEE Main 2019 (Online) 9th January Evening Slot

The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is :

(A) 3 (B) 2 (C) 4 (D) 5

Open solution

Q119. JEE Main 2018 (Online) 16th April Morning Slot

Let p, q and r be real numbers ($p \neq q$, $r \neq 0$), such that the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then the sum of squares of these roots is equal to :

(A) $\frac{p^2+q^2}{2}$ (B) $p^2 + q^2$ (C) $2(p^2 + q^2)$ (D) $p^2 + q^2 + r^2$

Open solution

Q120. JEE Main 2018 (Online) 16th April Morning Slot

If an angle A of a ΔABC satiesfies $5\cos A + 3 = 0$, then the roots of the quadratic equation, $9x^2 + 27x + 20 = 0$ are :

(A) secA, cotA (B) sinA, secA (C) secA, tanA (D) tanA, cosA

Open solution

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Q121. JEE Main 2018 (Offline)

Let S = { $x \in R : x \geq 0$ and

$2\left|\sqrt{x}-3\right|+\sqrt{x}\left(\sqrt{x}-6\right)+6=0\}$. Then S

(A) contains exactly four elements

(B) is an empty set

(C) contains exactly one element

(D) contains exactly two elements

Open solution

Q122. JEE Main 2018 (Online) 15th April Evening Slot

If f(x) is a quadratic expression such that f (1) + f (2) = 0, and -1 is a root of f (x) = 0, then the other root of f(x) = 0 is :

(A) $-\frac{5}{8}$ (B) $-\frac{8}{5}$ (C) $\frac{5}{8}$ (D) $\frac{8}{5}$

Open solution

Q123. JEE Main 2018 (Online) 15th April Morning Slot

If $\lambda \in$ **R** is such that the sum of the cubes of the roots of the equation,
$x^2 + (2-\lambda)x + (10-\lambda) = 0$ is minimum, then the magnitude of the difference of the roots of this equation is :

(A) $4\sqrt{2}$ (B) $2\sqrt{5}$ (C) $2\sqrt{7}$ (D) 20

Open solution

Q124. JEE Main 2018 (Online) 15th April Morning Slot

If tanA and tanB are the roots of the quadratic equation, $3x^2 - 10x - 25 = 0$, then the value of $3\sin^2(A+B) - 10\sin(A+B).\cos(A+B) - 25\cos^2(A+B)$ is :

(A) -10 (B) 10 (C) -25 (D) 25

Open solution

Q125. JEE Main 2017 (Online) 9th April Morning Slot

The sum of all the real values of x satisfying the equation
$2^{(x-1)(x^2+5x-50)} = 1$ is :

(A) 16 (B) 14 (C) -4 (D) -5

Open solution

Q126. JEE Main 2017 (Online) 8th April Morning Slot

Let p(x) be a quadratic polynomial such that p(0)=1. If p(x) leaves remainder 4 when divided by $x-1$ and it leaves remainder 6 when divided by x + 1; then :

(A) p(2) = 11 (B) p(2) = 19 (C) p(-2) = 19 (D) p(-2) = 11

Open solution

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Q127. JEE Main 2017 (Offline)

If for a positive integer n, the quadratic equation

$$x(x+1)+(x+1)(x+2)+\ldots+\left(x+\overline{n-1}\right)(x+n)=10n$$

has two consecutive integral solutions, then n is equal to :

(A) 9 (B) 10 (C) 11 (D) 12

Open solution

Q128. JEE Main 2016 (Online) 10th April Morning Slot

If x is a solution of the equation, $\sqrt{2x+1}-\sqrt{2x-1}=1,\ \left(x\geq\frac{1}{2}\right)$, then $\sqrt{4x^2-1}$ is equal to :

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) $2\sqrt{2}$

Open solution

Q129. JEE Main 2016 (Online) 9th April Morning Slot

If the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root different from −1, then $|b|$ is equal to :

(A) $\sqrt{2}$ (B) 2 (C) 3 (D) $\sqrt{3}$

Open solution

Q130. JEE Main 2016 (Offline)

The sum of all real values of x satisfying the equation $\left(x^2-5x+5\right)^{x^2+4x-60}=1$ is :

(A) 6 (B) 5 (C) 3 (D) -4

Open solution

Q131. JEE Main 2015 (Online) 11th April Morning Slot

If the two roots of the equation,

$(a-1)(x^4+x^2+1)+(a+1)(x^2+x+1)^2=0$

are real and distinct, then the set of all values of 'a' is :

(A) $\left(-\frac{1}{2},0\right)$ (B) $(-\infty,-2)\cup(2,\infty)$

(C) $\left(-\frac{1}{2},0\right)\cup\left(0,\frac{1}{2}\right)$ (D) $\left(0,\frac{1}{2}\right)$

Open solution

Q132. JEE Main 2015 (Online) 10th April Morning Slot

If $2+3i$ is one of the roots of the equation $2x^3-9x^2+kx-13=0$, $k\in\mathbf{R}$, then the real root of this equation :

(A) does not exist. (B) exists and is equal to $\frac{1}{2}$

(C) exists and is equal to $-\frac{1}{2}$. (D) exists and is equal to 1.

Open solution

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Q133. JEE Main 2015 (Offline)

Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to :

(A) 3 (B) -3 (C) 6 (D) -6

Open solution

Q134. JEE Main 2014 (Online) 19th April Morning Slot

The equation $\sqrt{3x^2 + x + 5}$ = x − 3, where x is real, has :

(A) no solution
(B) exactly one solution
(C) exactly two solutions
(D) exactly four solutions.

Open solution

Q135. JEE Main 2014 (Online) 19th April Morning Slot

Let f(n) = $\left[\frac{1}{3} + \frac{3n}{100}\right]n$, where [n] denotes the greatest integer less than or equal to n.

Then $\sum_{n=1}^{56} f(n)$ is equal to :

(A) 56 (B) 689 (C) 1287 (D) 1399

Open solution

Q136. JEE Main 2014 (Online) 19th April Morning Slot

If non-zero real numbers b and c are such that *min f(x) > max g(x),* where

$f(x) = x^2 + 2bx + 2c^2$ and
$g(x) = -x^2 - 2cx + b^2$ $(x \in R)$;

then $\left|\frac{c}{b}\right|$ lies in the interval :

(A) $\left(0, \frac{1}{2}\right)$ (B) $\left[\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$
(C) $\left[\frac{1}{\sqrt{2}}, \sqrt{2}\right]$ (D) $\left(\sqrt{2}, \infty\right)$

Open solution

Q137. JEE Main 2014 (Online) 12th April Morning Slot

The sum of the roots of the equation, $x^2 + |2x - 3| - 4 = 0$, is :

(A) 2 (B) -2 (C) $\sqrt{2}$ (D) $-\sqrt{2}$

Open solution

Q138. JEE Main 2014 (Online) 11th April Morning Slot

If α and β are roots of the equation,
$x^2 - 4\sqrt{2}kx + 2e^{4\ln k} - 1 = 0$
for some k, and $\alpha^2 + \beta^2 = 66$, then $\alpha^3 + \beta^3$ is equal to :

(A) $248\sqrt{2}$ (B) $280\sqrt{2}$ (C) $-32\sqrt{2}$ (D) $-280\sqrt{2}$

Open solution

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Q139. JEE Main 2014 (Online) 9th April Morning Slot

If equations $ax^2 + bx + c = 0$, $(a, b, c \in \mathbf{R}, a \neq 0)$ and $2x^2 + 3x + 4 = 0$ have a common root, then a : b : c equals :

(A) 1 : 2 : 3 (B) 2 : 3 : 4 (C) 4 : 3 : 2 (D) 3 : 2 : 1

Open solution

Q140. JEE Main 2014 (Online) 9th April Morning Slot

If $\frac{1}{\sqrt{\alpha}}$ and $\frac{1}{\sqrt{\beta}}$ are the roots of the equation, $ax^2 + bx + 1 = 0$ $(a \neq 0, a, b \in \mathbf{R})$, then the equation, $x(x + b^3) + (a^3 - 3abx) = 0$ has roots :

(A) $\alpha^{\frac{3}{2}}$ and $\beta^{\frac{3}{2}}$

(B) $\alpha\beta^{\frac{1}{2}}$ and $\alpha^{\frac{1}{2}}\beta$

(C) $\sqrt{\alpha\beta}$ and $\alpha\beta$

(D) $\alpha^{-\frac{3}{2}}$ and $\beta^{-\frac{3}{2}}$

Open solution

Q141. JEE Main 2014 (Offline)

If $a \in R$ and the equation $-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$ (where $[x]$ denotes the greater integer $\leq x$) has no integral solution, then all possible values of a lie in the interval :

(A) $(-2, -1)$ (B) $(-\infty, -2) \cup (2, \infty)$

(C) $(-1, 0) \cup (0, 1)$ (D) $(1, 2)$

Open solution

Q142. JEE Main 2014 (Offline)

Let α and β be the roots of equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q, r in A.P. and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is :

(A) $\frac{\sqrt{34}}{9}$ (B) $\frac{2\sqrt{13}}{9}$ (C) $\frac{\sqrt{61}}{9}$ (D) $\frac{2\sqrt{17}}{9}$

Open solution

Q143. JEE Main 2013 (Online) 25th April Morning Slot

If p and q are non-zero real numbers and $\alpha^3 + \beta^3$ = -p, $\alpha\beta$ = q, then a quadratic equation whose roots are $\frac{\alpha^2}{\beta}$, $\frac{\beta^2}{\alpha}$ is:

(A) $px^2 - qx + p^2 = 0$ (B) $qx^2 + px + q^2 = 0$

(C) $px^2 + qx + p^2 = 0$ (D) $qx^2 - px + q^2 = 0$

Open solution

Q144. JEE Main 2013 (Online) 23th April Morning Slot

The least integral value α of x such that $\frac{x-5}{x^2+5x-14} > 0$, satisfies:

(A) $\alpha^2 + 3\alpha - 4 = 0$ (B) $\alpha^2 - 5\alpha + 4 = 0$ (C) $\alpha^2 - 3\alpha + 6 = 0$ (D) $\alpha^2 + 5\alpha - 6 = 0$

Open solution

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Q145. JEE Main 2013 (Online) 22th April Morning Slot

If α and β are roots of the equation $x^2 + px + \frac{3p}{4} = 0$, such that $|\alpha - \beta| = \sqrt{10}$, then p belongs to the set

(A) {2, -5} (B) {-3, 2} (C) {-2, 5} (D) {3, -5}

Open solution

Q146. JEE Main 2013 (Online) 9th April Morning Slot

The values of $'a'$ for which one root of the equation $x^2 - (a+1)x + a^2 + a - 8 = 0$ exceeds 2 and the other is lesser than 2, are given by

(A) $3 < a < 10$ (B) $a \geq 10$ (C) $-2 < a < 3$ (D) $a \leq -2$

Open solution

Q147. JEE Main 2013 (Offline)

If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0, a, b, c \in R$, have a common root, then $a : b : c$ is

(A) $1 : 2 : 3$ (B) $3 : 2 : 1$ (C) $1 : 3 : 2$ (D) $3 : 1 : 2$

Open solution

Q148. AIEEE 2012 (Online) 26th May

If $a, b, c \in \mathrm{R}$ and 1 is a root of equation $ax^2 + bx + c = 0$, then the curve $y = 4ax^2 + 3bx + 2c,\ a \neq 0$ intersect x-axis at

(A) two distinct points whose coordinates are always rational numbers

(B) no point

(C) exactly two distinct points

(D) exactly one point

Open solution

Q149. AIEEE 2012 (Online) 19th May

Let $p, q, r \in R$ and $r > p > 0$. If the quadratic equation $px^2 + qx + r = 0$ has two complex roots α and β, then $|\alpha| + |\beta|$ is

(A) equal to 1 (B) less than 2 but not equal to 1

(C) greater than 2 (D) equal to 2

Open solution

Q150. AIEEE 2012 (Online) 12th May

If the sum of the square of the roots of the equation $x^2 - (\sin\alpha - 2)x - (1 + \sin\alpha) = 0$ is least, then α is equal to

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Open solution

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Q151. AIEEE 2012 (Online) 7th May

The value of K for which the equation $(K-2)x^2+8x+K+4=0$ has both roots real, distinct and negative is

(A) 6 (B) 3 (C) 4 (D) 1

Open solution

Q152. AIEEE 2012

The equation $e^{\sin x}-e^{-\sin x}-4=0$ has:

(A) infinite number of real roots (B) no real roots

(C) exactly one real root (D) exactly four real roots

Open solution

Q153. AIEEE 2010

If α and β are the roots of the equation $x^2-x+1=0$, then $\alpha^{2009}+\beta^{2009}=$

(A) -1 (B) 1 (C) 2 (D) -2

Open solution

Q154. AIEEE 2009

If the roots of the equation $bx^2+cx+a=0$ are imaginary, then for all real values of x, the expression $3b^2x^2+6bcx+2c^2$ is :

(A) less than $4ab$ (B) greater than $-4ab$ (C) less than $-4ab$ (D) greater than $4ab$

Open solution

Q155. AIEEE 2008

STATEMENT - 1 : For every natural number $n \geq 2$,

$$\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\ldots\ldots\ldots+\frac{1}{\sqrt{n}}>\sqrt{n}.$$

STATEMENT - 2 : For every natural number $n \geq 2$,,

$$\sqrt{n\,(n+1)}<n+1.$$

(A) Statement - 1 is false, Statement - 2 is true

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for statement - 1

(C) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement - 1

(D) Statement - 1 is true, Statement - 2 is false

Open solution

Q156. AIEEE 2008

The quadratic equations $x^2-6x+a=0$ and $x^2-cx+6=0$ have one root in common. The other roots of the first and second equations are integers in the ratio 4 : 3. Then the common root is

(A) 1 (B) 4 (C) 3 (D) 2

Open solution

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Q157. AIEEE 2007

If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is

(A) $(3, \infty)$ (B) $(-\infty, -3)$ (C) $(-3, 3)$ (D) $(-3, \infty)$

Open solution

Q158. AIEEE 2006

If the roots of the quadratic equation $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$, respectively, then the value of $2 + q - p$ is

(A) 2 (B) 3 (C) 0 (D) 1

Open solution

Q159. AIEEE 2006

All the values of m for which both roots of the equation $x^2 - 2mx + m^2 - 1 = 0$ are greater than -2 but less than 4, lie in the interval

(A) $-2 < m < 0$ (B) $m > 3$ (C) $-1 < m < 3$ (D) $1 < m < 4$

Open solution

Q160. AIEEE 2006

If x is real, the maximum value of $\frac{3x^2+9x+17}{3x^2+9x+7}$ is

(A) $\frac{1}{4}$ (B) 41 (C) 1 (D) $\frac{17}{7}$

Open solution

Q161. AIEEE 2005

In a triangle PQR, $\angle R = \frac{\pi}{2}$. $If\ \tan\left(\frac{P}{2}\right)$ and $\tan\left(\frac{Q}{2}\right)$ are the roots of $ax^2 + bx + c = 0,\ a \neq 0$ then

(A) $a = b + c$ (B) $c = a + b$ (C) $b = c$ (D) $b = a + c$

Open solution

Q162. AIEEE 2005

If both the roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then k lies in the interval

(A) $(5, 6]$ (B) $(6, \infty)$ (C) $(-\infty, 4)$ (D) $[4, 5]$

Open solution

Q163. AIEEE 2005

The value of a for which the sum of the squares of the roots of the equation $x^2 - (a-2)x - a - 1 = 0$ assumes the least value is

(A) 1 (B) 0 (C) 3 (D) 2

Open solution

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Q164. AIEEE 2005

If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals

(A) -2 (B) 3 (C) 2 (D) 1

Open solution

Q165. AIEEE 2004

Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are the roots of the quadratic equation

(A) $x^2 - 18x - 16 = 0$ (B) $x^2 - 18x + 16 = 0$ (C) $x^2 + 18x - 16 = 0$ (D) $x^2 + 18x + 16 = 0$

Open solution

Q166. AIEEE 2004

If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its roots are

(A) $-1, 2$ (B) $-1, 1$ (C) $0, -1$ (D) $0, 1$

Open solution

Q167. AIEEE 2004

If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of $'q'$ is

(A) 4 (B) 12 (C) 3 (D) $\frac{49}{4}$

Open solution

Q168. AIEEE 2003

If the sum of the roots of the quadratic equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{a}{c}$, $\frac{b}{a}$ and $\frac{c}{b}$ are in

(A) Arithmetic - Geometric Progression (B) Arithmetic Progression

(C) Geometric Progression (D) Harmonic Progression

Open solution

Q169. AIEEE 2003

The value of 'a' for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is

(A) $-\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $-\frac{2}{3}$ (D) $\frac{1}{3}$

Open solution

Q170. AIEEE 2003

The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$ is

(A) 3 (B) 2 (C) 4 (D) 1

Open solution

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Q171. AIEEE 2002

If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$ then the equation having α/β and β/α as its roots is

(A) $3x^2 - 19x + 3 = 0$ (B) $3x^2 + 19x - 3 = 0$ (C) $3x^2 - 19x - 3 = 0$ (D) $x^2 - 5x + 3 = 0$

Open solution

Q172. AIEEE 2002

Difference between the corresponding roots of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ is same and $a \neq b$, then

(A) $a + b + 4 = 0$ (B) $a + b - 4 = 0$ (C) $a - b - 4 = 0$ (D) $a - b + 4 = 0$

Open solution

Q173. AIEEE 2002

Product of real roots of equation $t^2x^2 + |x| + 9 = 0$

(A) is always positive (B) is always negative

(C) does not exist (D) none of these

Open solution

Q174. AIEEE 2002

If p and q are the roots of the equation $x^2 + px + q = 0$, then

(A) $p = 1,\ q = -2$ (B) $p = 0,\ q = 1$ (C) $p = -2,\ q = 0$ (D) $p = -2,\ q = 1$

Open solution

Q175. AIEEE 2002

If $a,\ b,\ c$ are distinct $+ve$ real numbers and $a^2 + b^2 + c^2 = 1$ then $ab + bc + ca$ is

(A) less than 1 (B) equal to 1 (C) greater than 1 (D) any real no.

Open solution

Numerical 39 questions

Q176. JEE Main 2025 (Online) 2nd April Evening Shift

If the set of all $\mathrm{a} \in \mathbf{R} - \{1\}$, for which the roots of the equation $(1 - \mathrm{a})x^2 + 2(\mathrm{a} - 3)x + 9 = 0$ are positive is $(-\infty, -\alpha] \cup [\beta, \gamma)$, then $2\alpha + \beta + \gamma$ is equal to .

Open solution

Q177. JEE Main 2025 (Online) 23rd January Morning Shift

If the equation $\mathrm{a}(\mathrm{b} - \mathrm{c})\mathrm{x}^2 + \mathrm{b}(\mathrm{c} - \mathrm{a})\mathrm{x} + \mathrm{c}(\mathrm{a} - \mathrm{b}) = 0$ has equal roots, where $\mathrm{a} + \mathrm{c} = 15$ and $\mathrm{b} = \frac{36}{5}$, then $a^2 + c^2$ is equal to _____

Open solution

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Q178. JEE Main 2024 (Online) 8th April Evening Shift

The number of distinct real roots of the equation $|x+1||x+3|-4|x+2|+5=0$, is _____

Open solution

Q179. JEE Main 2024 (Online) 6th April Evening Shift

Let α, β be roots of $x^2+\sqrt{2}x-8=0$. If $U_n=\alpha^n+\beta^n$, then $\frac{U_{10}+\sqrt{2}U_9}{2U_8}$ is equal to _____.

Open solution

Q180. JEE Main 2024 (Online) 6th April Morning Shift

Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4+8x^3-17x^2-12x+9=0$ and $\left(4+x_1^2\right)\left(4+x_2^2\right)\left(4+x_3^2\right)\left(4+x_4^2\right)=\frac{125}{16}m$. Then the value of m is _____.

Open solution

Q181. JEE Main 2024 (Online) 5th April Evening Shift

The number of real solutions of the equation $x|x+5|+2|x+7|-2=0$ is _____.

Open solution

Q182. JEE Main 2024 (Online) 5th April Morning Shift

The number of distinct real roots of the equation $|x||x+2|-5|x+1|-1=0$ is _____.

Open solution

Q183. JEE Main 2024 (Online) 31st January Evening Shift

Let a, b, c be the lengths of three sides of a triangle satisfying the condition $\left(a^2+b^2\right)x^2-2b(a+c)x+\left(b^2+c^2\right)=0$. If the set of all possible values of x is the interval (α, β), then $12\left(\alpha^2+\beta^2\right)$ is equal to _____.

Open solution

Q184. JEE Main 2024 (Online) 30th January Evening Shift

The number of real solutions of the equation $x\left(x^2+3|x|+5|x-1|+6|x-2|\right)=0$ is _____.

Open solution

Q185. JEE Main 2024 (Online) 30th January Morning Shift

Let $\alpha, \beta \in \mathbf{N}$ be roots of the equation $x^2-70x+\lambda=0$, where $\frac{\lambda}{2}, \frac{\lambda}{3} \notin \mathbf{N}$. If λ assumes the minimum possible value, then $\frac{(\sqrt{\alpha-1}+\sqrt{\beta-1})(\lambda+35)}{|\alpha-\beta|}$ is equal to :

Open solution

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Q186. JEE Main 2024 (Online) 29th January Evening Shift

Let the set $C = \{(x, y) \mid x^2 - 2^y = 2023, x, y \in \mathbb{N}\}$. Then $\sum_{(x,y)\in C} (x + y)$ is equal to _____.

Open solution

Q187. JEE Main 2023 (Online) 13th April Evening Shift

Let $[\alpha]$ denote the greatest integer $\leq \alpha$. Then $[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + \ldots + [\sqrt{120}]$ is equal to _____

Open solution

Q188. JEE Main 2023 (Online) 11th April Evening Shift

The number of points, where the curve $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$ cuts x-axis, is equal to _____.

Open solution

Q189. JEE Main 2023 (Online) 11th April Morning Shift

If a and b are the roots of the equation $x^2 - 7x - 1 = 0$, then the value of $\frac{a^{21}+b^{21}+a^{17}+b^{17}}{a^{19}+b^{19}}$ is equal to _____.

Open solution

Q190. JEE Main 2023 (Online) 8th April Evening Shift

Let m and n be the numbers of real roots of the quadratic equations $x^2 - 12x + [x] + 31 = 0$ and $x^2 - 5|x + 2| - 4 = 0$ respectively, where $[x]$ denotes the greatest integer $\leq x$. Then $m^2 + mn + n^2$ is equal to _____.

Open solution

Q191. JEE Main 2023 (Online) 30th January Evening Shift

If the value of real number $a > 0$ for which $x^2 - 5ax + 1 = 0$ and $x^2 - ax - 5 = 0$

have a common real root is $\frac{3}{\sqrt{2\beta}}$ then β is equal to _____.

Open solution

Q192. JEE Main 2023 (Online) 29th January Evening Shift

Let $\alpha_1, \alpha_2, \ldots, \alpha_7$ be the roots of the equation $x^7 + 3x^5 - 13x^3 - 15x = 0$ and $|\alpha_1| \geq |\alpha_2| \geq \ldots \geq |\alpha_7|$. Then $\alpha_1\alpha_2 - \alpha_3\alpha_4 + \alpha_5\alpha_6$ is equal to _____.

Open solution

Q193. JEE Main 2023 (Online) 25th January Evening Shift

Let $a \in \mathbb{R}$ and let α, β be the roots of the equation $x^2 + 60^{\frac{1}{4}}x + a = 0$. If $\alpha^4 + \beta^4 = -30$, then the product of all possible values of a is _____.

Open solution

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Q194. JEE Main 2023 (Online) 24th January Morning Shift

Let $\lambda \in \mathbb{R}$ and let the equation E be $|x|^2 - 2|x| + |\lambda - 3| = 0$. Then the largest element in the set S = $\{x + \lambda : x$ is an integer solution of E$\}$ is _____

Open solution

Q195. JEE Main 2022 (Online) 30th July Evening Shift

The number of solutions of the equation $|x^2 + 3x + 2| + |x + 5| - 2 = 0, x \in \mathbb{R}$, is _____.

Open solution

Q196. JEE Main 2022 (Online) 29th July Evening Shift

Let $\alpha, \beta(\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in \mathrm{N}$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to _____.

Open solution

Q197. JEE Main 2022 (Online) 28th July Morning Shift

The sum of all real values of x for which $\frac{3x^2 - 9x + 17}{x^2 + 3x + 10} = \frac{5x^2 - 7x + 19}{3x^2 + 5x + 12}$ is equal to _____.

Open solution

Q198. JEE Main 2022 (Online) 26th July Morning Shift

If for some $\mathrm{p}, \mathrm{q}, \mathrm{r} \in \mathbf{R}$, not all have same sign, one of the roots of the equation $\left(\mathrm{p}^2 + \mathrm{q}^2\right)x^2 - 2\mathrm{q}(\mathrm{p} + \mathrm{r})x + \mathrm{q}^2 + \mathrm{r}^2 = 0$ is also a root of the equation $x^2 + 2x - 8 = 0$, then $\frac{\mathrm{q}^2 + \mathrm{r}^2}{\mathrm{p}^2}$ is equal to _____,

Open solution

Q199. JEE Main 2022 (Online) 26th July Morning Shift

The number of distinct real roots of the equation $x^5\left(x^3 - x^2 - x + 1\right) + x\left(3x^3 - 4x^2 - 2x + 4\right) - 1 = 0$ is _____.

Open solution

Q200. JEE Main 2022 (Online) 28th June Morning Shift

The number of real solutions of the equation $e^{4x} + 4e^{3x} - 58e^{2x} + 4e^x + 1 = 0$ is _____.

Open solution

Q201. JEE Main 2022 (Online) 27th June Evening Shift

Let α, β be the roots of the equation $x^2 - 4\lambda x + 5 = 0$ and α, γ be the roots of the equation $x^2 - \left(3\sqrt{2} + 2\sqrt{3}\right)x + 7 + 3\lambda\sqrt{3} = 0, \lambda > 0$. If $\beta + \gamma = 3\sqrt{2}$, then $(\alpha + 2\beta + \gamma)^2$ is equal to _____.

Open solution

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Q202. JEE Main 2022 (Online) 27th June Morning Shift

If the sum of all the roots of the equation

$e^{2x} - 11e^{x} - 45e^{-x} + \frac{81}{2} = 0$ is $\log_e p$, then p is equal to _____.

Open solution

Q203. JEE Main 2022 (Online) 26th June Evening Shift

Let p and q be two real numbers such that p + q = 3 and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p} + \frac{1}{q}\right)^{-2}$ is equal to _____.

Open solution

Q204. JEE Main 2022 (Online) 26th June Morning Shift

The sum of the cubes of all the roots of the equation

$x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$ is _____.

Open solution

Q205. JEE Main 2021 (Online) 1st September Evening Shift

Let f(x) be a polynomial of degree 3 such that
$f(k) = -\frac{2}{k}$ for k = 2, 3, 4, 5. Then the value of $52 - 10f(10)$ is equal to :

Open solution

Q206. JEE Main 2021 (Online) 26th August Evening Shift

Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation $x^2 - x + 2\lambda = 0$, and α and γ are the roots of equation $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to _____.

Open solution

Q207. JEE Main 2021 (Online) 26th August Morning Shift

The sum of all integral values of k ($k \neq 0$) for which the equation $\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$ in x has no real roots, is _____.

Open solution

Q208. JEE Main 2021 (Online) 27th July Evening Shift

The number of real roots of the equation $e^{4x} - e^{3x} - 4e^{2x} - e^{x} + 1 = 0$ is equal to _____.

Open solution

Q209. JEE Main 2021 (Online) 25th July Evening Shift

If a + b + c = 1, ab + bc + ca = 2 and abc = 3, then the value of $a^4 + b^4 + c^4$ is equal to _____.

Open solution

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Q210. JEE Main 2021 (Online) 25th July Morning Shift

If α, β are roots of the equation $x^2 + 5(\sqrt{2})x + 10 = 0$, $\alpha > \beta$ and $P_n = \alpha^n - \beta^n$ for each positive integer n, then the value of $\left(\frac{P_{17}P_{20}+5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19}+5\sqrt{2}P_{18}^2}\right)$ is equal to _____.

Open solution

Q211. JEE Main 2021 (Online) 26th February Evening Shift

Let α and β be two real numbers such that $\alpha + \beta = 1$ and $\alpha\beta = -1$. Let $p_n = (\alpha)^n + (\beta)^n$, $p_{n-1} = 11$ and $p_{n+1} = 29$ for some integer $n \geq 1$. Then, the value of p_n^2 is _____.

Open solution

Q212. JEE Main 2021 (Online) 26th February Morning Shift

The sum of 162th power of the roots of the equation $x^3 - 2x^2 + 2x - 1 = 0$ is _____.

Open solution

Q213. JEE Main 2021 (Online) 24th February Evening Shift

The number of the real roots of the equation $(x+1)^2 + |x-5| = \frac{27}{4}$ is _____.

Open solution

Q214. JEE Main 2020 (Online) 8th January Morning Slot

The least positive value of 'a' for which the equation

$2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **B**	3. **B**	4. **D**	5. **C**
6. **B**	7. **D**	8. **A**	9. **A**	10. **B**
11. **A**	12. **A**	13. **D**	14. **B**	15. **C**
16. **C**	17. **D**	18. **A**	19. **A**	20. **A**
21. **C**	22. **A**	23. **A**	24. **D**	25. **C**
26. **A**	27. **A**	28. **D**	29. **C**	30. **A**
31. **B**	32. **A**	33. **B**	34. **A**	35. **D**
36. **C**	37. **A**	38. **B**	39. **B**	40. **D**
41. **A**	42. **D**	43. **B**	44. **C**	45. **B**
46. **C**	47. **B**	48. **A**	49. **B**	50. **B**
51. **C**	52. **C**	53. **D**	54. **B**	55. **B**
56. **C**	57. **B**	58. **B**	59. **A**	60. **A**
61. **A**	62. **B**	63. **B**	64. **B**	65. **B**
66. **D**	67. **B**	68. **A**	69. **B**	70. **D**
71. **A**	72. **A**	73. **C**	74. **B**	75. **A**
76. **A**	77. **D**	78. **C**	79. **A**	80. **A**
81. **B**	82. **B**	83. **C**	84. **C**	85. **A**
86. **C**	87. **C**	88. **C**	89. **D**	90. **D**
91. **A**	92. **D**	93. **C**	94. **C**	95. **A**
96. **D**	97. **A**	98. **C**	99. **C**	100. **D**
101. **B**	102. **C**	103. **B**	104. **A**	105. **C**
106. **C**	107. **A**	108. **B**	109. **D**	110. **A**
111. **B**	112. **B**	113. **B**	114. **B**	115. **A**
116. **D**	117. **B**	118. **A**	119. **B**	120. **C**
121. **D**	122. **D**	123. **B**	124. **C**	125. **C**
126. **C**	127. **C**	128. **A**	129. **D**	130. **C**
131. **C**	132. **B**	133. **A**	134. **A**	135. **D**
136. **D**	137. **C**	138. **B**	139. **B**	140. **A**
141. **C**	142. **B**	143. **B**	144. **D**	145. **C**
146. **C**	147. **A**	148. **C**	149. **C**	150. **D**

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151. **B**	152. **B**	153. **B**	154. **B**	155. **B**
156. **D**	157. **C**	158. **B**	159. **C**	160. **B**
161. **B**	162. **C**	163. **A**	164. **D**	165. **B**
166. **C**	167. **D**	168. **D**	169. **B**	170. **C**
171. **A**	172. **A**	173. **A**	174. **A**	175. **A**

Numerical

176. **7**	177. **117**	178. **2**	179. **4**	180. **221**
181. **3**	182. **3**	183. **36**	184. **1**	185. **60**
186. **46**	187. **825**	188. **2**	189. **51**	190. **9**
191. **13**	192. **9**	193. **45**	194. **5**	195. **0**
196. **16**	197. **6**	198. **272**	199. **3**	200. **2**
201. **98**	202. **45**	203. **4**	204. **36**	205. **26**
206. **18**	207. **66**	208. **2**	209. **13**	210. **1**
211. **324**	212. **3**	213. **2**	214. **8**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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